

Carolyn Yu

RESEARCH ASSISTANT · HCI · USER EXPERIENCE

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Education

National Chengchi University - Visual Information Processing Lab

Taipei, Taiwan

M.S. IN COMPUTER SCIENCE (WITH GPA 4.0/4.0)

2015 - 2017

- Thesis title: Construction and Optimization of Feature Descriptor Based on Dynamic Local Intensity Order Relations of Pixel Group

National Chengchi University

Taipei, Taiwan

B.S. IN COMPUTER SCIENCE (WITH GPA 3.87/4.0)

2011 - 2015

B.S. IN DIGITAL CONTENT AND TECHNOLOGIES

Publication

WalkingVibe: Reducing Virtual Reality Sickness and Improving Realism While Walking in VR Using Unobtrusive Head-Mounted Vibrotactile Feedback

ACM CHI '20

YI-HAO PENG, **CAROLYN YU**, SHI-HONG LIU, CHUNG-WEI WANG, PAUL TAELE, NENG-HAO YU, AND MIKE Y. CHEN

April. 2020

In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (Acceptance rate: 24.3%)

BusMyFriend: Designing a Bus Reservation Service for People With Visual Impairments in Taipei City

ACM DIS'20

CAROLYN YU, YI LEE, TSAI-YUAN HUANG, WEI-AN HSIEH, SHAO-YU LEE, I-HUI YEH, GANG KU LIN, NENG-HAO YU, HSIEN-HUI TANG, AND YUNG-JU CHANG

July. 2020

In Companion Publication of the 2020 ACM Designing Interactive Systems Conference (Acceptance rate: 25%)

Construction and Optimization of Feature Descriptor Based on Dynamic Local Intensity Order Relations of Pixel Group

ICIAR'19

WEN-HUNG LIAO, **CAROLYN YU**, YI-CHIEH WU

Aug. 2019

In Proceedings of the 2019 International Conference on Image Analysis and Recognition

Industrial and Academic Experience

Innovation User Interface Lab, Department of Design, National Taiwan University of Science and Technology (NTUST)

Taipei, Taiwan

RESEARCH ASSISTANT

2018 - Present

- Proposed design guidelines of Auditory User Interface and Graphical User Interface for people with visual impairments.
- Developed high fidelity prototypes which provides realistic experience on bus-reservation mobile App for user testing in the field.
- Planned and conducted user research to study the usability of prototypes, which improved 25.3% of System Usability Scale Score in accordance with users' feedback in the final version.
- Designed and developed a wearable haptic device that significantly reduces the VR sickness mean score from 22.2 to 9.2.
- Designed human factor experiments to define the patterns of haptic feedback and analyzed the data of questionnaires and interviews to prove the proposed hypothesis.
- Developed an eye-tracking system in Unity including data recording, analysis, and visualization to understand the relationship between VR sickness and the user's eye-movement behavior in the virtual environment.

Visual Information Processing Lab, Department of Computer Science, National Chengchi University

Taipei, Taiwan

RESEARCH ASSISTANT

2015 - 2017

- Proposed a novel approach to model image pattern features via pairwise intensity relation in pixel groups, which outperforms the state-of-the-art method by 1.5%.
- Developed a behavior analysis software via object recognition techniques with temporal information processing for analyzing high school student behaviors on 3D modeling software.
- Developed and conducted multiple educational applications, including motion-sensing games, low-cost 3D portrait construction software, and interactive behavior analysis experiments for middle school students via infrared cameras such as Kinect, Leap Motion, RealSense and HoloLens.

3D Modeling & Printing Class (Novel Applications and Key Technologies Development of 3D Printing in Elementary and High School Education)

Taipei, Taiwan

TEACHING ASSISTANT

2017

- Designed and customized several educational maker challenges for all-aged students to practice laser cutting and 3D printing techniques.
- Devised various curriculum targeting different age groups of students (8-16 y/o) to establish 3D geometrical concepts via 3D model construction.

Programming Studio (Computer Science Program for High School Students)

Taipei, Taiwan

TEACHING ASSISTANT

2016

- Guided students through the classwork and designed challenges for advanced ones.

Human Computer Interaction

Taipei, Taiwan

TEACHING ASSISTANT

2016

- Instructed students to build interactive prototypes with Arduino.

LEADERG Inc.

Taipei, Taiwan

FRONT END ENGINEER INTERN

2015

- Optimized the usability of the customized electronic element drawing interface, a drawing board with multiple layers of specific shapes built-in, by adjusting the pop-up view in an organized layout with PHP.

Trend Micro Department of Safesync for Enterprise

Taipei, Taiwan

QUALITY ASSURANCE INTERN

2013-2014

- Verified that the system meets the client's needs via a set of testing cases converted from the requirements and design documents and ensured the efficiency of the procedure.

Honors & Awards

2018	1st Place , Heart x Hack Hackathon 2018	Taiwan
2018	2nd Place , Bayer Taiwan Innovation Award 2018	Taiwan
2017	Finalist , NASA Hackathon SPACE APPS CHALLENGE 2017 Taiwan	Taiwan
2016	Excellent Merit Scholarship , College of Science, National Chengchi University	Taiwan
2012	3rd Place , Department of Computer Science, College of Science Programming Contest	Taiwan

Exhibitions & Talks

2020	EyeBus , Design For Taiwan 4th Social Design Exhibition	Taiwan
2017	Remind , The 3rd Annual Conference of Taiwan Association of Computer-Human Interaction	Taiwan
2016	3D Maker Learning in School , Maker Faire Taipei 2016	Taiwan
2015	Dear Women, I'mperfect , Aether: NCCU DCT 3rd Graduation Exhibition	Taiwan

Extracurricular Activity

21st ACM International Conference on Mobile Human-Computer Interaction

Taipei, Taiwan

STUDENT VOLUNTEER

2019

Student Association of Computer Science Department

Taipei, Taiwan

MINISTER OF ARTISTIC DESIGN DEPARTMENT

2012 - 2013

NCCU Computer Science Summer Camp

Taipei, Taiwan

VICE COORDINATOR

2013

Voluntary Service Camp for Native Villages of Changhua High School Alumni Association

Changhua, Taiwan

STUDENT VOLUNTEER & MINISTER OF ARTISTIC DESIGN DEPARTMENT

2012

Skills

Programming	Python, C/C++, Java, Unity, HTML, CSS, JavaScript, Swift
Image Processing	OpenCV
Prototype	Arduino, Processing, Blender, Adobe Creative Suite, Sketch
Tools	Latex, Firebase, SPSS